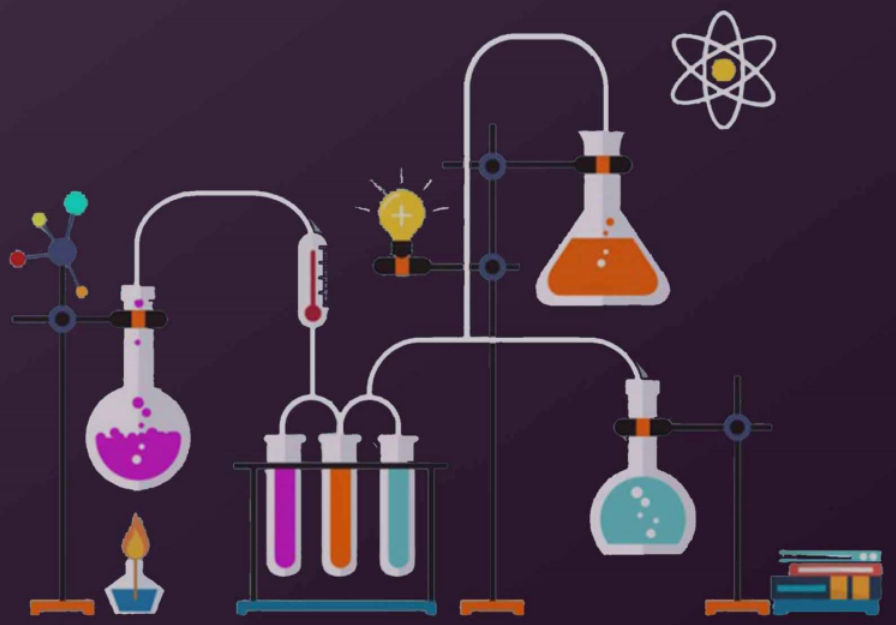
CHEMISTRY

Mole Concept



SHEET

eSaral.com

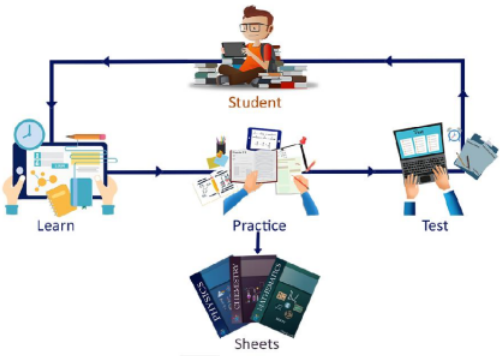
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 8
5.	Exercise 2	Page - 15
6.	Exercise 2A	Page - 19
7.	Answer Key	Page - 25

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Mole Concept	1-6	1-2		
2	Mole Basics	7-26	3-4, 20-27, 40,44-45	1-5	1,26
3	Empirical and Molecular Formula	27-33	5-6,28,29	6-9	2,3,15
4	Stoichiometry - Part 1	34-44	7-9	10-12	4
5	Stoichiometry - Part 2	45-52	10,38	13	5-7
6	Stoichiometry - Part 3	53-58	11-12,30, 31,39	14-15	21,35
7	Concentration Terms - Part 1	59-70	13-15, 46-47	16-22	8,16, 17,18, 23,27,28
8	Concentration Terms - Part 2	71-81	41	23-36	24,29,30
9	Concentration Terms - Part 3 (Typical Concentration Terms)	82-84	32-33, 48,49	37	9,31,36
10	Eudiometry	85-88	16,17, 34-37,42	38-40	10,22, 32-34,37
11	ACC		18,19,43, 50-59	41	11-14 19, 20,25

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Mole Concept

Q1. How much oxygen will add the 36 g carbon to give 132 g carbon dioxide assuming complete combustion of carbon?

(A) 16 g (B) 36 g

(C) 56 g (D) 96 g

Q2. A sample of Na_2CO_3 has Na = 43.5%, C = 11.5% and O = 45%. If the law of constant proportions is true, then the mass of Na in 5g of Na_2CO_3 from another source will be?

(A) 2.051 g (B) 0.2175 g

(C) 0.02175 g (D) 20.0 g

Q3. Which of the following compound show law of multiple proportion ?

(A) H_2O, Na_2O (B) CO_2, SO_2

(C) H_2O, H_2O_2 (D) H_2O, I_2O_5

Q4. What volume of oxygen would be required for the complete combustion of 100 litres of ethane according to the following equation $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$.

(A) 700L (B) 500L

(C) 350L (D) 200L

Q5. One of the following combination illustrate law of reciprocal proportions :

(A) N_2O_3, N_2O_4, N_2O_5

(B) NaCl, NaBr, NaI

(C) CS_2, CO_2, SO_2

(D) PH_3, P_2O_3, P_2O_5

Q6. Iron forms two oxides, in first oxide 56 gram. Iron is found to be combined with 16 gram oxygen and in second oxide 112 gram iron is found to be combined with 48 gram oxygen. This data satisfy the law of -

(A) Conservation of mass

(B) Reciprocal proportion

(C) Multiple proportion

(D) Combining volume

Mole Basics

Q7. If mass of one atom is 3.32×10^{-23} g, then calculate number of nucleons (neutrons and protons) present in 2 atoms of the element :

(A) 40 (B) 20

(C) 10 (D) 40 N_A

Q8. Mass of one atom of the element A is 3.9854×10^{-23} g. How many atoms are contained in 1 g of the element A ?

(A) 2.509×10^{23} (B) 6.022×10^{23}

(C) 12.044×10^{23} (D) 2.509×10^{22}

Q9. How many moles are present in 12×10^{23} molecules of ethyl alcohol :-

(A) 4 mole (B) 8 mole

(C) 12 mole (D) 2 mole

Q10. One atom of an element x weigh 6.643×10^{-23} g. Number of moles in 20 kg is :

(A) 4 (B) 40

(C) 100 (D) 500

Q11. 16 g of SO_x occupies 5.6 litre at STP. Assuming ideal gas nature, The value of x is :

(A) 1 (B) 2

(C) 3 (D) 4

Q12. The total number of electrons in one molecule of carbon dioxide is?

(A) 66 (B) 88

(C) 44 (D) 22

Q13. The number of g-molecules of oxygen in 6.02×10^{24} CO molecules is ?

(A) 1 g-molecule (B) 0.5 g-molecule
(C) 5 g-molecule (D) 10 g-molecule

Q14. Rearrange the following (I to IV) in the order of increasing masses.

(I) 0.5 mole of O_3
(II) 0.5 gm molecule of Nitrogen
(III) 3.011×10^{23} molecules of O_2
(IV) 11.35 L of CO_2 at STP.

(A) IV < III < II < I (B) II < III < IV < I
(C) III < II < I < IV (D) I < II < III < IV

Q15. Haemoglobin contain 2.8% Fe. The minimum molecular mass of haemoglobin is?

(A) 1000 (B) 2000
(C) 200 (D) 100

Q16. If the vapour density of gas is 17 then gas would be?

(A) NH_3 (B) CO_2
(C) H_2O_2 vapour (D) Both (A) & (C)

Q17. Average atomic mass of magnesium is 24.31 a.m.u. This magnesium is composed of 79 mole % of ^{24}Mg and remaining 21 mole % of ^{25}Mg and ^{26}Mg. Calculate mole % of ^{26}Mg.

(A) 10 (B) 11
(C) 15 (D) 16

Q18. The average molecular mass of a mixture of gas containing nitrogen and carbon dioxide is 36. The mixture contain 280 g of nitrogen, therefore, the amount of CO_2 present in the mixture is-

(A) 440 g (B) 44 g
(C) 0.1mole (D) 880 g

Q19. An iodized salt contains 0.5 % of NaI. A person consumes 3 g of salt everyday. The number of iodide ions going into his body everyday is -

(A) 10^{-4} (B) 6.02×10^{-4}
(C) 6.02×10^{19} (D) 6.02×10^{23}

Q20. A gaseous mixture contains 70% N_2 and a 30% unknown gas, by volume. If the average molecular weight of gaseous mixture is 37.60, the molecular weight of unknown gas is?

(A) 42.2 (B) 60
(C) 40 (D) 50

Q21. If 1 kg Zinc cost Rs. 100 & 1 kg Silver cost Rs. 100. To buy 1 mol of Zinc & 1 mol of Silver which will be cheaper ?

(A) Zinc is cheaper
(B) Silver is cheaper
(C) Both will cost same
(D) Can't predict

Q22. Which of the following represents the formula of a substance which contains about 25.92% nitrogen and 74.07% oxygen.

(A) N_2O (B) NO
(C) NO_2 (D) N_2O_5

Q23. The chloride of a metal contains 71% chlorine by weight and the vapour density of it is 50. The atomic mass of the metal will be (valency of metal is 2) -

(A) 29 (B) 58
(C) 35.5 (D) 71

Q24. In an ionic compound moles ratio of cation to anion is 1 : 2. If atomic masses of metal and non-metal respectively are 138 and 19, then correct statement is -

(A) molecular mass of compound is 176
(B) formula mass of compound is 176
(C) formula mass of compound is 157
(D) molecular mass of compound is 157

Q25. Two oxides of a metal contain 50% and 40% metal M respectively. If the formula of the first oxide is MO_2, the formula of the second oxide will be -

(A) MO_2 (B) MO_3
(C) M_2O (D) M_2O_5

Q26. Which of the following has the least mass ?
(A) 2 g atoms of nitrogen
(B) 3×10^{23} atoms of carbon
(C) 1 mol of sulphur
(D) 7.0 g of Ag

Empirical and Molecular Formula

Q27. Empirical formula of glucose is ?
(A) $C_6H_{12}O_6$ (B) $C_3H_6O_3$
(C) $C_2H_4O_2$ (D) CH_2O

Q28. A hydrocarbon contains 80% of carbon, then the hydrocarbon is?
(A) CH_4 (B) C_2H_4
(C) C_2H_6 (D) C_2H_2

Q29. A compound contains 69.5% oxygen and 30.5% nitrogen and its molecular weight is 92. The formula of that compound is?
(A) N_2O (B) NO_2
(C) N_2O_4 (D) N_2O_5

Q30. A carbon compound containing carbon and oxygen has molar mass equal to 288. On analysis it is found to contain 50% by mass of each element. Therefore molecular formula of the compound is-
(A) $C_{12}O_9$ (B) C_4O_3
(C) C_3O_4 (D) C_9O_{12}

Q31. A given sample of pure compound contains 9.81 g of Zn, 1.8×10^{23} atoms of chromium, and 0.60 mol of oxygen atoms. what is the simplest formula -
(A) $ZnCr_2O_7$ (B) $ZnCr_2O_4$
(C) $ZnCrO_4$ (D) $ZnCrO_6$

Q32. When 3.2 g S is vapourized at 450°C and 723 mm pressure, the vapours occupy a volume of 780 ml. What is the molecular formula of S vapours under these conditions -
(A) S_2 (B) S_4
(C) S_6 (D) S_8

Q33. 74 g of a sample on complete combustion gives 132 g CO_2 and 54 g of H_2O. The molecular formula of the compound may be
(A) C_5H_{12} (B) $C_4H_{10}O$
(C) $C_3H_6O_2$ (D) $C_3H_7O_2$

Stoichiometry Part - 1

Q34. A chemical equation is balanced according to?
(A) Conservation of number of atoms
(B) Conservation of.No. of molecule
(C) Conservation of.No. of Volume
(D) None of these

Q35. For $2H_2(g) + O_2(g) \rightarrow 2H_2O$ (l) 2 litre $H_2(g)$ will combine with $O_2(g)$
(A) 2 lit. (B) 1 lit.
(C) 4 lit. (D) 6 lit.

Q36. How many gram pure $KMnO_4$ should be heated with H_2SO_4 to produce 48 g O_2 ?
$KMnO_4 + H_2SO_4 \rightarrow K_2SO_4 + MnSO_4 + H_2O + O_2$
(A) 189.6 g (B) 122.5 g
(C) 158 g (D) 316 g

Q37. 18.625 g of KCl is produced by the decomposition of $KClO_4$ as follows :
$KClO_4(s) \longrightarrow KCl(s) + 2O_2(g)$
then, the volume of oxygen produced at STP will be?
(A) 11.2 L (B) 22.4 L
(C) 0.5 L (D) 16 L

Q38. 12 moles of each A and B allowed to react as given $3A + 2B \longrightarrow C + \frac{1}{2}D$. If 60 g of D is produced then, calculate the atomic mass of D :-
(A) 30 (B) 45 (C) 60 (D) 15

Q39. 3 g of Mg is burnt in a closed vessel containing 3 g of oxygen. The weight of excess reactant left is -
(A) 0.5 g of oxygen (B) 1.0 g of oxygen
(C) 1.0 g of Mg (D) 0.5 g of Mg

Q40. One mole of potassium chlorate ($KClO_3$) is thermally decomposed and excess of aluminium is burn in the gaseous product. How many mol of aluminium oxide (Al_2O_3) are formed:-

(A) 1 (B) 1.5 (C) 2 (D) 3

Q41. A sample of calcium carbonate is 80% pure. 25 g of this sample is treated with excess of HCl. How much volume of CO_2 will be obtained at N.T.P. ?

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

(A) 2.24 L (B) 5.6 L

(C) 11.2 L (D) 4.48 L

Q42. How much ammonia will be formed from 2.95 g of acetamide at STP.

$CH_3CONH_2 + NaOH \rightarrow CH_3COONa + NH_3$

(A) 1120 ml (B) 2240 ml

(C) 4000 ml (D) 4200 ml

Q43. 200g of calcium carbonate on being strongly heated the volume of CO_2 gas evolved at N.T.P. will be?

(A) 22.4 L (B) 44.8 L

(C) 11.2 L (D) None of these

Q44. If 0.5 moles of $BaCl_2$ is mixed with 0.2 mole of Na_3PO_4, the maximum number of mole of $Ba_3(PO_4)_2$ that can be formed is?

(A) 0.7 (B) 0.5 (C) 0.3 (D) 0.1

Stoichiometry - Part -2

Q45. If 9 moles of O_2 and 14 moles of N_2 are placed in a container and allowed to react according to the equation $3O_2 + 2N_2 \longrightarrow 2N_2O_3$. The reaction proceeds until 3 moles of O_2 remain, how many moles of N_2O_3 are present at that instant ?

(A) 6 (B) 3

(C) 4 (D) 12

Q46. Phosphine (PH_3) decomposes to produce P_4 (g) and H_2 (g). What would be the change in volume when 100 ml of PH_3 (g) is completely decomposed ?

(A) 50 ml (B) 500 ml

(C) 75 ml (D) 250 ml

Q47. 1.5 g of NaI reacts with 1.6 g of Br_2 to produce 0.412 g of NaBr, then determine the percent yield of the reaction ?

[Atomic weight of Na = 23, Br = 80, I = 127]

$2NaI + Br_2 \longrightarrow 2NaBr + I_2$

(A) 30% (B) 40%

(C) 50% (D) 60%

Q48. 2.2 gram of impure sodium chloride is treated with excess of $AgNO_3$ solution. 2.11 g of silver chloride ppt are formed. What is the % purity of sodium chloride :-

(A) 39.09 (B) 93.09

(C) 45 (D) 58

Q49. Calculate the volume of air required for the combustion of 6 litre of propane (C_3H_8)

(A) 100 litre (B) 30 litre

(C) 150 litre (D) 125 litre

Q50. The following sequence of reactions may be used to extract zinc from its sulphide :-

[Atomic weight of Zn = 65, S = 32]

$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$

$ZnO + C \longrightarrow Zn + CO$

How many tons of Zn can be obtained form 32.33 tons of ZnS, assuming that the yield is 75% for each reaction ?

(A) 5 (B) 100

(C) 16.2 (D) 16.4

Q51. NX is produced by the following steps of reactions :

$M + X_2 \rightarrow MX_2$

$M X_2 + X_2 \rightarrow M_3X_8$

$M_3X_8 + N_2CO_3 \rightarrow NX + CO_2 + M_3O_4$

How much M (Metal) is consumed to produce 206 g of NX

(Given at. wt. of M = 56, N = 23, X = 80)

(A) $\frac{7}{4}$ g (B) $\frac{14}{3}$

(C) 56 (D) 42

Q52. Sodium nitrate, heated in the presence of an excess of H_2, forms H_2O according to two step process:

$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$

$2H_2 + O_2 \longrightarrow 2H_2O$

How many grams of $NaNO_3$ are required to form 9 g of H_2O ?

(A) 21.3 (B) 42.5

(C) 69.0 (D) 85.0

Stoichiometry - Part -3

Q53. A gaseous mixture of propane and butane of total 3 moles on complete combustion produces 11.0 moles CO_2 under standard conditions of temperature and pressure.

The ratio of moles of butane to propane is :

(A) 1 : 2 (B) 2 : 1

(C) 3 : 2 (D) 3 : 1

Q54. Consider a 1.0000 g sample containing 75% potassium sulfate (FW 174.25) and 25% MSO_4. The sample is dissolved and the sulfate is precipated as $BaSO_4$ (FW 233.39). If the $BaSO_4$ ppt weighs 1.4900, what is the atomic weight of M^{2+} in MSO_4 ?

(A) 24.12 (B) 10

(C) 20 (D) 35

Q55. What will be volume of the mixture after the reaction

(g)NH_3 (1L) + (g)HCl (1.5L) $\rightarrow$ NH_4Cl (s)

(A) 1.5 L (B) 0.5 L

(C)1 L (D) 0 L

Q56. A granulated sample of aircraft alloy (Al, Mg, Cu) weighing 8.72g was first treated with alkali and then with very dilute HCl, leaving a residue. The residue after alkali boiling weighed 2.10g and the acid insoluble residue weighed 0.69g. Which of the following is correct

(A) Weight % of Al = 75.9%

(B) Weight % of Cu = 7.9%

(C) Weight % of Mg = 16.2 %

(D) All the above

Q57. Calculate persentage mole composition of CO_2 in final mixture when 24 g of carbon reacts with 80 g of oxygen gas (assume 100% reaction).

(A) 100 (B) 66.67

(C) 33.33 (D) None of these

Q58. 39 g of an alloy of aluminium and magnesium when heated with excess of dil. HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected at 0°C has a volume of 44.8 litres at 1 atm pressure. Calculate the composition of the alloy by moles.

(A) 45.6% (B) 35.6%

(C) 66.6% (D) 50%

Concentration Terms Part-1

Q59. Which one of the following modes of expressing concentration of solution is independent of temperature :

(A) Molarity (B) Molality

(C) % w/v (D) Grams per litre

Q60. Mole fraction of $C_3H_5(OH)_3$ in a solution of 36 g of water and 46 g of glycerine ($C_3H_8O_3$) is

(A) 0.46 (B) 0.36

(C) 0.20 (D) 0.40

Q61. Mole fraction of methanol in its aqueous solution is 0.5. The concentration of solution in terms of percent by mass of methanol is -

(A) 36 (B) 50

(C) 64 (D) 72

Q62. 8 g NaOH is dissolved in one litre of solution, its molarity is ?

(A) 0.8 M (B) 0.4 M

(C) 0.2 M (D) 0.1 M

Q63. The molality of a solution prepared by dissolving 125 mL of pure methanol ($d = 0.8\ g\ mL^{-1}$) in 375 g of ethanol is :-

(A) 10.8 (B) 11.2

(C) 8.33 (D) 10.4

Q64. If 18 g of glucose is present in 1000 g of solvent, the solution is said to be :
(A) 1 molar (B) 0.01 molal
(C) 0.5 molar (D) 0.1 molal

Q65. The concentration of H_2SO_4 in a solution which has a density 1.2 g/ml and mass percent of H_2SO_4 is 9.8% is :
(A) 9.9 M (B) 1.2 M
(C) 0.6 M (D) 1.8 M

Q66. The mole fraction of a given sample of I_2 in C_6H_6 is 0.2. The molality of I_2 in C_6H_6 is :-
(A) 0.32 (B) 0.032
(C) 0.48 (D) 3.2

Q67. The number of mole of NaOH present in 180 g of 0.2 molal aqueous solution is ?
(A) 0.2 (B) 0.036
(C) 0.003 (D) 0.01

Q68. The mass of carbon present in 36.8g of potassium ferrocyanide (Mol. mass = 368) is 12g.
(A) 1.1 g (B) 2.2 g
(C) 3.3 g (D) 4.4 g

Q69. A 20% solution of KOH (density = 1.02g/ml) has molarity = 3.64.
(A) 1.33 (B) 3.64
(C) 3.31 (D) 4.41

Q70. The molarity of 20% (W/W) solution of sulphuric acid is 2.55 M. The density of the solution is ?
(A) 1.25 g cm^{-3} (B) 0.125 g L^{-1}
(C) 2.55 g cm^{-3} (D) Unpredictable

Concentration Part -2

Q71. Two solutions of a substance (non - electrolyte) are mixed in the following manner 480 mL of 1.5 M of first solution with 520 mL of 1.2 M of second solution. The molarity of final solution is :
(A) 1.20 M (B) 1.50 M
(C) 1.344 M (D) 2.70 M

Q72. Equal volumes of 10% (v/v) of HCl is mixed with 10% (v/v) NaOH solution. If density of pure NaOH is 1.5 times that of pure HCl then the resultant solution be.
(A) basic
(B) neutral
(C) acidic
(D) can't be predicted.

Q73. Phosphine [$PH_3(g)$] decomposes to produce vapours of phosphorus (P_4) and H_2 gas. What will be the change in volume when 100 mL of phosphine is decomposed ?
(A) +50 mL (B) 500 mL
(C) +75 mL (D) –500 mL

Q74. Assuming complete precipitation of AgCl, calculate the sum of the molar concentration of all the ions if 2 lit of 2M Ag_2SO_4 is mixed with 4 lit of 1 M NaCl solution is :
(A) 4M (B) 2M
(C) 3 M (D) 2.5 M

Q75. How much $BaCl_2$ would be needed to make 250 ml of a solution having the same concentration of Cl^- as one containing 3.78g NaCl per 100 ml ? (Ba = 137)
(A) 16.8 g (B) 15.8
(C) 35.7 (D) 10.6

Q76. Find out the volume of 98% w/w H_2SO_4 (density = 1.8 g/ml), must be diluted to prepare 12.5 litres of 2.5 M sulphuric acid solution.
(A) 1.536 L (B) 1.736 L
(C) 1.336 L (D) 1.25 L

Q77. Which concentration unit is not affected by temperature :-
(A) Normality (B) Formality
(C) Molarity (D) Molality

Q78. Molarity and % w/W of a solution of NaOH is 5 and 15. The density of solution is ?
(A) 1 (B) 1.33
(C) 1.67 (D) 2.33

Q79. Two solution labelled as 3 M HCl and 1M HCl are mixed in the ratio of x : y by volume and the molarity of resulting solution is 1.5 M. What is the molarity of the resulting solution if they are mixed in the ratio y : x by volume ?

(A) 4M (B) 3M

(C) 2M (D) 2.5M

Q80. The volume of water which must be added to a mixture of 350 cm^3 of 6M HCl and 650 ml of 3M HCl to get a resulting solution of 3M concentration is ?

(A) 75 ml (B) 150 ml

(C) 100 ml (D) 350 ml

Q81. A mixture has 18 g water and 414 g ethanol. The mole fraction of water in mixture is ?

(A) 0.1 (B) 0.4

(C) 0.7 (D) 0.9

Concentration Part -3

Q82. An oleum sample is labelled as 118 %, Calculate mass of H_2SO_4 in 100 g oleum sample.

(A) 20 (B) 30

(C) 40 (D) 60

Q83. If 50 g oleum sample rated as 118% is mixed with 18 g water, then the correct option is

(A) The resulting solution contains 18 g of water and 118 g H_2SO_4

(B) The resulting solution contains 9 g water and 59 g H_2SO_4

(C) The resulting solution contains only 118 g pure H_2SO_4

(D) The resulting solution contains 68 g of pure H_2SO_4

Q84. One litre of a sample of hard water contain 4.44 mg $CaCl_2$ and 1.9 mg of $MgCl_2$. What is the hardness in terms of ppm of $CaCO_3$?

(A) 2 ppm (B) 3 ppm

(C) 4 ppm (D) 6 ppm

Eudiometry

Q85. A definite amount of gaseous hydrocarbon was burnt with just sufficient amount of O_2. The volume of all reactants was 600 ml, after the explosion the volume of the products [CO_2(g) and H_2O(g)] was found to be 700 ml under the similar conditions. The molecular formula of the compound is -

(A) C_3H_8 (B) C_3H_6

(C) C_3H_4 (D) C_4H_{10}

Q86. The % by volume of C_4H_{10} in a gaseous mixture of C_4H_{10}, CH_4 and CO is 40. When 200 ml of the mixture is burnt in excess of O_2. Find volume (in ml) of CO_2 produced.

(A) 220 (B) 340

(C) 440 (D) 560

Q87. 10 ml of a compound containing 'N' and 'O' is mixed with 30 ml of H_2 to produce H_2O (*l*) and 10 ml of N_2 (g). Molecular formula of compound if both reactants reacts completely, is?

(A) N_2O (B) NO_2

(C) N_2O_3 (D) N_2O_5

Q88. 100 ml of CH_4 and C_2H_2 were exploded with excess of O_2. After explosion and cooling, the mixture was treated with KOH, where a reduction of 165 ml was observed. Therefore the composition of the mixture is -

(A) CH_4 = 35 ml ; C_2H_2 = 65 ml

(B) CH_4 = 65 ml ; C_2H_2 = 35 ml

(C) CH_4 = 75 ml ; C_2H_2 = 25 ml

(D) CH_4 = 25 ml ; C_2H_2 = 75 ml

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. What is true regarding Dalton's atomic theory -
(A) All atom of same element possess same mass & same size.
(B) Atoms are destructible.
(C) Atoms combine in definite ratio.
(D) It is the smallest particle which can take part in chemical reaction.

Q2. Two flask of equal capacity contain NH_3 & SO_2 gases respectively, are kept under similar conditions of temperature & pressure. Select the correct option on the basis of given information :
(A) More moles are present in flask contain NH_3
(B) Flask of SO_2 has more mass
(C) Flask of NH_3 has more number of atoms.
(D) Both flask contain same number of molecules of NH_3 & SO_2 respectively

Q3. It is known that atom contain protons, neutrons and electrons. If the mass of neutron is assumed to half of its original value where as that of electron is assumed to be twice of its original value then the atomic mass of $^{14}_{6}C$ will be?
(A) same (B) 25% less
(C) 50% less (D) 28.5% less

Q4. 10 mole of A_2B_3 contains 100 g A atom & 60 g B atoms. Choose the correct statements -
(A) Molecular weight of A_2B_3 is equal to 16
(B) Atomic weight of A is equal to 5
(C) Weight of one atom of B is equal to 2
(D) Atomic weight of B is equal to 6

Q5. The molecular formula of a compound is $A_2B_3C_4$. If 5 g of the compound contains 0.6 g of 'A', 2.4 g of 'B' and 0.04 mole of C-atoms, then the correct information(s) is/are?
(A) The atomic mass of 'A' is 30
(B) The atomic mass of 'B' is 60
(C) The atomic mass of 'C' is 50
(D) The molecular mass of compound is 160

Q6. 1 mol BaF_2 + 2 mol H_2SO_4 → resulting mixture will be neutralised by ?
(A) 1 mol of KOH
(B) 2 mol of $Ca(OH)_2$
(C) 4 mol KOH
(D) 2 mol of KOH

Q7. Which of the following option(s) is heavier than 1g-molecule oxygen -
(A) 12g of O_3
(B) 1g-molecule O_3
(C) 4g-atom of hydrogen
(D) 1.12 litre of H_2O at 4°C and 1 atm

Q8. When $FeCl_3$ is ignited in an atmosphere of pure oxygen, this reaction takes place :
$4FeCl_3(s) + 3O_2(g) \longrightarrow 2Fe_2O_3(s) + 6Cl_2(g)$
If 3 moles of $FeCl_3$ are ignited in the presence of 2 moles of O_2 gas. Which of the following statements regarding to the given reaction is/are correct ?
(A) 0.33 mole $FeCl_3$ remains unreacted
(B) 0.67 mole $FeCl_3$ remains unreacted
(C) 1.33 moles of Fe_2O_3 are formed
(D) 0.50 mole O_2 remains unreacted

Q9. A mixture of C_4H_8 and C_2H_4 was completely burnt in excess of oxygen yielding equal volumes of CO_2 and steam. Calculate the percentage of the compounds in the original mixture :
(A) 25% C_4H_8 and 75% C_2H_4
(B) 30% C_4H_8 and 70% C_2H_4
(C) 75% C_4H_8 and 25% C_2H_4
(D) 50% C_4H_8 and 50% C_2H_4

Q10. 100 ml of 0.5M - $Ba(OH)_2$ solution is mixed with 200 ml of 0.2M- H_3PO_3 solution and the mixture is diluted to 1000 ml. The final concentration of ions present in the solution is?

[The salt formed is completely insoluble in water and H^+ and OH^- ions in solution are only due to H_3PO_3 and $Ba(OH_2)$]

(A) $[Ba^{2+}] = 0.0$ M

(B) $[Ba^{2+}] = 0.01$ M

(C) $[PO_3^{3-}] = 0.02$ M

(D) $[HPO_3^{2-}] = 0.0$ M

Q11. If mass of precipitate obtained is 25 g in each of the following decomposition reaction per 100g reactant. Which of the following reaction has yield greater than 50%.

(A) $2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

(B) $NaNO_3 \longrightarrow NaNO_2 + \frac{1}{2}O_2$

(C) $Ca(NO_3)_2 \longrightarrow CaO + 2NO_2 + \frac{1}{2}O_2$

[Ca = 40]

(D) $PbCO_3 \longrightarrow PbO + CO_2$ [Pb = 208]

Q12. 11.2 g of mixture of MCl (volatile) and NaCl gave 28.7 g of white ppt with excess of $AgNO_3$ solution. 11.2 g of same mixture on heating gave a gas that on passing into $AgNO_3$ solution gave 14.35 g of white ppt. Hence ?

(A) Ionic mass of M+ is 18

(B) Mixture has equal mol fraction of MCl and NaCl

(C) MCl and NaCl are in 1 : 2 molar ratio

(D) Ionic mass of M+ is 10

Q13. The mole fraction of NaCl in aqueous solution is 0.2. The solution is?

(A) 13.9 m

(B) Mole fraction of H_2O is 0.8

(C) acidic in nature

(D) neutral

Q14. 90 g glucose is dissolved in 800 g water to get a solution of density $\frac{89}{80}$ m / ml. The correct concentration of solution is /are?

(A) $\frac{90}{8}$% (w/w) (B) $\frac{90}{8}$% (w/v)

(C) 0.625 m (D) 0.625 M

Q15. An oleum sample has SO_3 and H_2SO_4 in 2 : 3 mass ratio. Select the correct statement(s)

(A) % labelling of sample is 109%

(B) % labelling of sample is 118%

(C) If 9gm H_2O is added to 200 gm above sample new labelling would be 104.5 %

(D) If 9gm H_2O is added to 200 gm above sample new labelling would be 104.3 %

Q16. Two gases A and B which react according to the equation $aA_{(g)} + bB_{(g)} \longrightarrow cC_{(g)} + dD_{(g)}$ to give two gases C and D are taken (amount not known) in an Eudiometer tube (operating at a constant Pressure and temperature) to cause the above.

If on causing the reaction there is no volume change observed then which of the following statement is/are correct.

(A) (a + b) = (c + d)

(B) Average molecular mass may increase or decrease if either of A or B is present in limited amount.

(C) Vapour Density of the mixture will remain same throughout the course of reaction.

(D) Total moles of all the component of mixture will change.

Q17. 4 g of C_3H_8 and 14 g of O_2 are allowed to react maximum possible extent to forms only CO & H_2O. In final gaseous mixture which of the given relation is correct-

(A) $\frac{n_{CO}}{n_{O_2}} = \frac{16}{7}$ (B) $\%w_{CO} = \frac{200}{3}$

(C) $W_{CO} = 7.636$ g (D) $W_{CO} = 14$ g

Q18. Sulphur molecule exists under various condition as S_8, S_6, S_4 and S_2. Which of the following statement(s) is (are) incorrect :

(A) Mass of one mole of each of these is same

(B) Number of molecules in one mole of each of these is same

(C) Number of atoms in one mole of each of these is same

(D) 1g of each contain same no. of atoms

Q19. In the formation reaction of NH_3 from N_2 & H_2, 140 g of N_2 and 40 g H_2 were mixed. Select the option which is/are incorrect.

(A) Maximum mass of NH_3 which can be formed is 180 g.

(B) If % yield of reaction is 80%, then H_2 consumed will be 32 g.

(C) Some N_2 (g) will be left after the reaction

(D) If NH_3 formed is 85 g then % yield will be 50%.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q.20 to 22

Three isotopes of Mg is available in nature.

They are $_{12}Mg^{24}$, $_{12}Mg^{25}$, $_{12}Mg^{26}$

The relative abundance of the isotopes are as follows :

Isotopes	Relative abundance
$_{12}Mg^{24}$	80%
$_{12}Mg^{25}$	5%
$_{12}Mg^{26}$	15%

Q20. Which isotope has maximum number of atoms per gram of each ?

(A) $_{12}Mg^{24}$ (B) $_{12}Mg^{25}$

(C) $_{12}Mg^{26}$ (D) equal

Q21. Number of neutrons are highest in

(A) $_{12}Mg^{24}$ (B) $_{12}Mg^{25}$

(C) $_{12}Mg^{26}$ (D) equal

Q22. Average atomic mass of Mg is approximately equal to -

(A) 25.0 (B) 24.3

(C) 25.2 (D) 25.8

Paragraph for Q.23 to 27

Atoms and molecules are so small in size that they cannot be counted individually. If we take a small sample of any substance it contains very large number of atoms and molecules. In order to overcome this problem the concept of mole was introduced. In this concept number of atoms or molecules is related with mass of the substance. Number of atoms presents in 12 gram carbon (C-12 isotope) is experimentally determined and found to be equal to 6.022×10^{23}. This number is know as one mole, it is also known as Avogadro's number and is denoted by symbol N_0.

Q23. Mass of 3.01×10^{20} molecule of H_2O is?

(A) 18g (B) 18mg (C) 9g (D) 9mg

Q24. Total number of atoms in 4.4 grams CO_2 is?

(A) 1.806×10^{23} (B) 0.602×10^{23}

(C) 6.02×10^{23} (D) 12.04×10^{23}

Q25. Number of moles of oxygen molecules in 16 mg is?

(A) 0.5 (B) 1

(C) 5×10^{-4} (D) 2

Q26. Which one has minimum number of atoms ?

(A) 8 gram oxygen

(B) 1 gram hydrogen

(C) 7 gram nitrogen

(D) 1.5 gram helium

Q27. Maximum number of atoms belongs to (at STP) -

(A) 5.0 litres CO_2

(B) 5.6 litres N_2

(C) 6.0 litres of O_2

(D) all have equal atoms

Paragraph for Q.28 to 29

An unknown compound "A" composed of manganese and oxygen, has 36.78% oxygen by weight. If 1.7 g of "A" is heated in steam of hydrogen gas, water is formed in association with 1.38 g of "B" (Mn_xO_y).

Q28. The formula of A can be -

(A) MnO (B) MnO_2

(C) Mn_2O_3 (D) Mn_2O_4

Q29. Weight of water formed in this reaction -

(A) 0.35 g (B) 2.34 g

(C) 1.17 g (D) 0.117 g

Paragraph for Q.30 to 31

A crystaline hydrated salt on being rendered anhydrous loses 45.6% of its weight.

The percentage composition of anhydrous salt is : Al = 10.5%, K = 15.1%, S = 24.8% and oxygen = 49.6% [Molar mass are : Al = 27, K = 39, S = 32].

Answering following question based on these information.

Q30. What is the empirical formula of the salt -

(A) $K_2AlS_2O_7$ (B) $K_2Al_2S_2O_7$

(C) $KAlS_2O_8$ (D) $K_3AlS_2O_{12}$

Q31. What is the empirical formula of the hydrated salt?

(A) $K_2AlS_2O_7.10H_2O$

(B) $K_2Al_2S_2O_7.16H_2O$

(C) $K_3AlS_2O_{12}.8H_2O$

(D) $KAlS_2O_8.12H_2O$

Paragraph for Q.32 to 33

2 litre of 9.8 % w/w H_2SO_4 (d = 1.5 gm/ml) solution is mixed with 3 litre of 1 M KOH solution.

Q32. The number of moles H_2SO_4 added are

(A) 1 (B) 2

(C) 3 (D) 0.5

Q33. The concentration of H^+ if solution is acidic or concentration of OH^- if solution is basic in the final solution is -

(A) 0 (B) $\frac{3}{10}$

(C) $\frac{3}{5}$ (D) $\frac{2}{5}$

Paragraph for Q.34 to 37

A 10 ml mixture of N_2, an Alkane & O_2 undergo combustion in Eudiometry tube. There was contraction of 2 ml, when residual gases are passed through KOH. To the remaining mixture comprising of only one gas excess H_2 was added & after reaction the gas produced is absorbed by water, causing a reduction in volume of 8 ml.

Q34. Gas produced after introduction of H_2 in the mixture?

(A) H_2O (B) CH_4

(C) CO_2 (D) NH_3

Q35. Volume of N_2 present in the mixture?

(A) 2 ml (B) 4 ml

(C) 6 ml (D) 8 ml

Q36. Volume of O_2 remained after the first combustion?

(A) 4 ml (B) 2 ml

(C) 0 (D) 8 ml

Q37. Identify the hydrocarbon.

(A) CH_4 (B) C_2H_6

(C) C_3H_8 (D) C_4H_{10}

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q38. Match the following, and select the correct code :-

List-I	List-II
(P) Same number of atoms	(1) 4.25 g NH_3 and 4.5 g H_2O
(Q) Same number of molecules	(2) 2.20 g CO_2 and 0.90 g H_2O
(R) Same numbers of atoms as well as molecules	(3) 4.0 g CH_3Cl and 05.0 g NH_3
(S) Different numbers of atoms as well as molecules	(4) 3.2 g O_2 and 2.80 g CO

	P	Q	R	S
(A)	4	1	3	2
(B)	2	4	3	1
(C)	4	1	2	3
(D)	3	4	2	1

Q39. **List- I** **List- II**

(P) $\underset{1\,g}{2H_2} + \underset{1\,g}{O_2} \rightarrow 2H_2O$ (1) 0.56 g of CaO

(Q) $\underset{1\,g}{N_2} + \underset{1\,g}{3H_2} \rightarrow 2NH_3$ (2) 1.333 g

(R) $\underset{1\,g}{CaCO_3} \rightarrow CaO + CO_2$ (3) 1.125 g

(S) $\underset{1\,g}{2H_2} + \underset{1\,g}{C} \rightarrow CH_4$ (4) 1.214 g

	P	Q	R	S
(A)	3	4	1	2
(B)	2	4	3	1
(C)	4	1	2	3
(D)	3	4	2	1

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q40. Match the Column :

Column-I	Column-II
(A) 16 g of CH_4	(P) 0.5 mole
(B) 1 g of H_2	(Q) 5 N_A atoms
(C) 22 g of CO_2	(R) 11.2 litre at STP
(D) 9 g of H_2O	(S) 0.5 × N_A atoms
	(T) 1.5 N_A atoms

Q41. **Column-I** **Column-II**

(A) Concentration which is dimensionless (P) ppm

(B) Concentration which remain same on varying temperature (Q) Molality

(C) In a dilute aqueous solution let mole fraction of solute is x, (R) $\frac{\text{mol}}{\text{mol}}$ %

therefore $\frac{x \times 10^3}{(1-x) \times 18}$ represent

(D) In dilute aqueous solutions mg/L concentration is approximately (S) $\frac{\text{wt.}}{\text{wt.}}$%

Q42. Gaseous alkane (C_nH_{2n+2}) exploded with oxygen. Ratio of the mole of O_2 for complete combustion to the mole of CO_2 formed is given in column-I & in column II formula is given.

Column-I	**Column-II**
(P) 7 : 4	(1) C_3H_8
(Q) 2 : 1	(2) C_4H_{10}
(R) 5 : 3	(3) C_2H_6
(S) 13 : 8	(4) CH_4

Code:

	P	**Q**	**R**	**S**
(A)	3	2	1	2
(B)	2	4	1	3
(C)	3	4	1	2
(D)	2	3	1	4

Q43. **Column-I** **Column-II**

(Reactions with 2 moles A and 5 moles B) **(Characteristics)**

(A) A + 2B → products (P) 'A' is limiting reagent

(B) A + 3B → products (Q) 'B' is limiting reagent

(C) 2A + 5B → products (R) During chemical reaction, mole ratio of reactants will not change

(D) A + B → products (S) 3 moles of excess reactant is left after complete reaction

(T) 1 moles of excess reactant is left after complete reaction

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q44. If a protein has 0.07% Fe (M = 56) by weight as the only metal, its molar mass would be at least M × 10^4 g. Here M is?

Q45. The abundance of three isotopes of oxygen are as follows

% of O^{16} = 90%

% of O^{17} + % of O^{18} = 10%

Assume at. mass same as mass no. Find out % of O^{17} and O^{18}, if the avg. atomic mass is 16.12.

Q46. A sample of oleum is labelled as $\frac{745}{7}\%$.

The molality of oleum is (mole of SO_3 per kg of H_2SO_4).

Q47. Density of a solution containing 13% by mass of sulphuric acid is 0.98 g/mL. Then molarity of solution will be ?

Q48. 500 ml of 2 M NaCl solution was mixed with 200 ml of 2 M NaCl solution. Calculate the final volume and molarity of NaCl in final solution if final solution has density 1.5 g/ml.

Q49. A 6.90 M solution of KOH in water contains 30% by mass of KOH. What is density of solution in g/ml.

Q50. Find out the volume of 98% w/w H_2SO_4 (density = 1.8 gm/ ml), must be diluted to prepare 12.5 litres of 2.5 M sulphuric acid solution.

Q51. 10 ml of a mixture of CO, CH_4 and N_2 exploded with excess of oxygen gave a contraction of 6.5 ml. There was a further contraction of 7 ml, when the residual gas treated with KOH. Volume of CO, CH_4 and N_2 respectively is ?

Q52. 20 ml of a mixture of C_2H_2 and CO was exploded with 30 ml of oxygen. The gases after the reaction had a volume of 34 ml. On treatment with KOH, 8 ml of oxygen remained. Calculate the volume of C_2H_2 in the mixture.

Q53. 10 ml of CO is mixed with 25 ml air having 20% O_2 by volume. What would be the final volume if none of CO and O_2 is left after the reaction?

Q54. 20% surface sites have adsorbed N_2. On heating N_2 gas evolved from sites and were collected at 0.001 atm and 298 K in a container of volume is 2.46 cm^3. Density of surface sites is $6.023\times10^{14}/cm^2$ and surface area is 1000 cm^2, find out the no. of surface sites occupied per molecule of N_2.

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q55. **Statement-1** : Molality of pure ethanol is lesser than pure water.

Statement -2 : As density of ethanol is lesser than density of water.

[Given : $d_{ethanol}$ = 0.789 gm/ml ;

d_{water} = 1 gm/ml]

Q56. **Statement-1 :** Vapour density of sulphur vapour relative to oxygen is 2 because sulphur atom is twice as heavy as that of O atom.

Statement-2 : Vapour density depends upon the molecular mass of the substance.

Q57. **Statement-1 :** Equal volumes of all the gases contain equal number of atoms.

Statement-2 : Atom is the smallest particle which takes part in chemical reactions.

SUBJECTIVE TYPE

Q58. A mixture is prepared by mixing 10 g H_2SO_4 and 40 g SO_3 calculate,

(a) mole fraction of H_2SO_4

(b) % labelling of oleum

Q59. A mixture of three gases an alkane (general formula C_nH_{2n+2}), an alkene (general formula C_xH_{2x}) and O_2 was subjected to sparking to cause combustion of both the hydrocarbon at 127°C. After the reaction three gases were present and none of the hydrocarbon remained. On passing the gases through KOH (absorb CO_2), an increment in mass of KOH solution by 132 g was observed. The remaining gases were passed over white anhydrous $CuSO_4$ and the weight of blue hydrated $CuSO_4$ crystals was found to be 72 g more than that of white anhydrous $CuSO_4$. Given that initially total 10 moles of the three gases were taken and **moles of alkane and alkene were equal** and if molecular mass of alkene – molecular mass of alkane=12 i.e. ($M_{alkene} - M_{alkane} = 12$), then answer the following questions. (Show calculations)

(a) Which three gases are remained after the combustion reactions

(b) What are the number of moles of product gases.

(c) What is the molecular formula of the two hydrocarbon.

(d) What is the number of moles of each of the two hydrocarbons and O_2 gas taken initially.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Number of electrons present in 3.6 mg of NH_4^+ are?

(A) 1.2×10^{21} (B) 1.2×10^{20}

(C) 1.2×10^{22} (D) 2×10^{-3}

Q2. The total number of valence electrons in 4.2g of N^{-3} ion is?

(A) 2.1 N_A (B) 4.2 N_A

(C) 1.6 N_A (D) 2.4 N_A

Q3. The mass of sulphuric acid needed for dissolving 3 g magnesium carbonate is?

(A) 3.5 g (B) 7.0 g

(C) 1.7 g (D) 17 g

Q4. An organic compound on analysis was found to contain 0.032% of sulphur. The molecular mass of the compound, if its molecule contains two sulphur atoms is?

(A) 200 (B) 2000

(C) 20000 (D) 200000

Q5. 56 g of red hot iron is treated with 36 mL water. The weight of Fe_3O_4 produced would be (Fe = 56).

(A) 77.33 g (B) 11.6 g

(C) 232 g (D) 58 g

Q6. Four grams of hydrocarbon (C_xH_y) on complete combustion gave 12 gram of CO_2. What is the empirical formula of the hydrocarbon :-

(A) CH_3 (B) C_4H_9

(C) CH (D) C_3H_8

Q7. In a closed vessel 50 ml of A_2B_3 completely reacts with 200 ml of C_2 according to the following equation

$2A_2B_3(g) + 5C_2(g) \rightarrow 3C_3B_2(g) + CA_4(g)$

The composition of gaseous mixture in the system will be?

(A) 100 ml C_2, 50 ml C_3B_2, 50 ml CA_4

(B) 25 ml C_2, 75 ml C_3B_2, 25 ml CA_4

(C) 75 ml C_2, 75 ml C_3B_2, 25 ml CA_4

(D) 10 ml C_2, 25 ml C_3B_2, 100 ml CA_4

Q8. What is the empirical formula of vanadium oxide, if 2.74 g of the metal oxide contains 1.53 g of metal ?

(A) V_2O_3 (B) VO

(C) V_2O_5 (D) V_2O_7

Q9. 1.61 g of $Na_2SO_4.10H_2O$ contains same number of oxygen atoms as present in

(A) 0.98 g H_2SO_4 (B) 0.08 g SO_3

(C) 1.78 g $H_2S_2O_7$ (D) 0.05 g $CaCO_3$

Q10. 2.8 litres of oxygen at NTP is equivalent to :-

(A) $\frac{1}{4}$ mole of O_2 (B) $\frac{1}{8}$ moles of O_2

(C) 4 g of O_2 (D) Both (B) & (C)

Q11. Rearrange the following (I to IV) in the order of increasing masses :-

(I) 0.5 mole of O_3

(II) 0.5 g atom of oxygen

(III) 3.01×10^{23} molecules of O_2

(IV) 5.6 litre of CO_2 at S.T.P.

(A) II < IV < III < I

(B) II < I < IV < III

(C) IV < II < III < I

(D) I < II < III < IV

Q12. The chief ore of Zn is the sulphide, ZnS. The ore is concentrated by froth floatation process and then heated in air to convert ZnS to ZnO.

$2ZnS + 3O_2 \xrightarrow{80\%} 2ZnO + 2SO_2$

$ZnO + H_2SO_4 \xrightarrow{100\%} ZnSO_4 + H_2O$

$2\,ZnSO_4 + 2H_2O \xrightarrow{80\%} 2Zn + 2H_2SO_4 + O_2$

The number of moles of ZnS required for producing 2 moles of Zn will be?

(A) 3.125 (B) 2

(C) 2.125 (D) 4

Q13. 10 ml of propane are mixed with 70 ml of oxygen and exploded.

What will be the volume of residual gases after explosions :-

(A) 55 ml (B) 40 ml

(C) 50 ml (D) 35 ml

Q14. Sulphur trioxide is prepared by the following two reactions

$S_{8(s)} + 8O_{2(g)} \rightarrow 8SO_{2(g)}$

$2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$

How many grams of SO_3 are produced from 1 mole S_8

(A) 320 g (B) 640 g

(C) 960 g (D) 420g

Q15. When 280 g of ethylene polymerises to polyethylene according to the equation.

$n(CH_2{=}CH_2) \longrightarrow -(CH_2{-}CH_2)_n-$

The weight and mole of polyethylene formed will be?

(A) 280, 10n (B) $\frac{280}{n}$, n

(C) $\frac{280}{n}$, 280 (D) 280, $\frac{10}{n}$

Q16. If the vapour density of gas is 8.5. Then calculate the weight of 2.24L of gas at STP :-

(A) 17 g (B) 1.7 g

(C) 2.24 g (D) 8.5 g

Q17. 2.0 molar solution is obtained, when 0.5 mole solute is dissolved in :-

(A) 250 ml solvent

(B) 250 g solvent

(C) 250 ml solution

(D) 1000 ml solvent

Q18. Assuming that petrol is C_8H_{18} and has density 0.8 g/ml. 1.425 litre of petrol on complete combustion will consume :

(A) 80 mole of O_2

(B) 170 mole of O_2

(C) 200 mole of O_2

(D) 125 mole of O_2

Q19. The molality of 1 L solution of 93% H_2SO_4 (W/V) having density 1.84 g/mL is?

(A) 1.043 m (B) 0.143 m

(C) 10.43 m (D) 5.215 m

Q20. When 1L of CO_2 is heated with graphite, the volume of the gases collected is 1.5 L. Calculate the number of moles of CO produced at STP.

(A) $\frac{1}{11.2}$ (B) $\frac{28}{22.4}$

(C) $\frac{1}{22.4}$ (D) $\frac{14}{22.4}$

Q21. A solution is prepared by mixing 100 mL each of CH_3OH(d = 0.8 g/mL) and C_2H_5OH (d = 1 g/mL). If M_1 is molarity of solution having CH_3OH solute in C_2H_5OH solvent and M_2 is molarity of solution having C_2H_5OH solute in CH_3OH solvent. Then :-

(A) $M_1 < M_2$ (B) $M_1 > M_2$

(C) $M_1 = M_2$ (D) $M_1 + M_2 > 50$

Q22. What volume of 75% alcohol by weight (d = 0.80 g/cm^3) must be used to prepare 150 cm^3 of 30% alcohol by weight (d = 0.90 g/cm^3) :-

(A) 67.5 mL (B) 56.25 mL

(C) 44.44 mL (D) 32.5 mL

Q23. 1.12 mL of a gas was produced at STP by the action of 4.12 mg of alcohol (ROH) with CH_3MgI. The molecular mass of ROH is?

(A) 16.0 (B) 41.2

(C) 82.4 (D) 156.0

Q24. What is the molarity of 10% by weight NaOH solution. If density of solution is 1.2g/ml :-

(A) 3M (B) 0.3M

(C) 1.5M (D) 2 M

Q25. A solution weighing 'a' gm has molality b. Find the molecular mass of solute if the mass of solute is 'c' gm :-

(A) $\frac{c}{b}\times\frac{1000}{(a-c)}$ (B) $\frac{b}{a}\times\frac{1000}{(a-b)}$

(C) $\frac{b}{c}\times\frac{1000}{(a-c)}$ (D) $\frac{c}{a}\times\frac{1000}{(b-a)}$

Q26. SO_2Cl_2 (sulphuryl chloride) reacts with water to given a mixture of H_2SO_4 and HCl. What volume of 0.2 M $Ba(OH)_2$ is needed to completely neutralize 25 mL of 0.2M SO_2Cl_2 solution :

(A) 25 mL (B) 50 mL

(C) 100 mL (D) 200 mL

Q27. Molarity of solution when 40 g NaOH is present 10 L solution would be?

(A) 0.1 M (B) 4 M

(C) 1 M (D) 0.4 M

Q28. What is the molarity of SO_4^{2-} ion in aqueous solution that contain 34.2 ppm of $Al_2(SO_4)_3$? (Assume complete dissociation and density of solution 1 g/mL) :-

(A) 3×10^{-4} M (B) 2×10^{-4} M

(C) 10^{-4} M (D) None of these

Q29. Mole fraction of water in 20% w/w H_2O_2 solution is ?

(A) $\frac{77}{68}$ (B) $\frac{68}{77}$ (C) $\frac{20}{80}$ (D) $\frac{9}{77}$

Q30. The density of a solution containing 13% by mass of H_2SO_4 is 1.09 gm/ml. The molarity of the solution is?

(A) 14.45 M (B) 1.445 M
(C) 1.09 M (D) None

Q31. 3.0 molal NaOH solution has a density of 1.110 g/ml. The molarity of the solution is?

(A) 2.94 (B) 3.25 (C) 3.64 (D) 1.25

Q32. The volume of water that must be added to a mixture of 250 ml of 6 M HCl and 750 ml of 2 M HCl to obtain 2.5 M solution is?

(A) 75 ml (B) 150 ml
(C) 100 ml (D) 200 ml

Q33. The relation between molarity (M) and molality (m) is given by (ρ = density of solution, M_1 = Molecular weight of solute) :-

(A) $m = \dfrac{1000M}{1000\rho - M_1}$

(B) $m = \dfrac{1000\rho M}{1000\rho - MM_1}$

(C) $m = \dfrac{1000MM}{1000\rho - MM_1}$

(D) $m = \dfrac{1000M}{1000\rho - MM_1}$

Q34. 1380 g Ag_2CO_3 sample decompose to produce sufficient oxygen required for complete combustion of C_2H_2 which produces 11.2 L of CO_2 at STP after combustion, then determine the percent purity of Ag_2CO_3 sample ?
[Atomic weight of Ag = 108]

$$Ag_2CO_3(s) \rightarrow 2Ag(s) + CO_2(g) + \frac{1}{2}O_2(g)$$

$$C_2H_2(g) + \frac{5}{2}O_2(g) \rightarrow 2CO_2(g) + H_2O(l)$$

(A) 25% (B) 50%
(C) 75% (D) 35%

Q35. How much volume of 3.0 M H_2SO_4 is required for the preparation of 1.0 litre of 1.0 M solution?

(A) 300 ml (B) 320 ml
(C) 333.3 ml (D) 350.0 ml

Q36. Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions molarity of the two solutions will be ?

(A) Equal
(B) That of NaCl will be less than that of KCl
(C) That of NaCl will be more than that of KCl Solution
(D) That of NaCl will be half of that of KCl solution

Q37. 12.5g of fuming H_2SO_4 (labelled as 112%) is mixed with 100 lit water. Molar concentration of H^+ in resultant solution is ?
[Note : Assume that H_2SO_4 dissociate completely and there is no change in volume on mixing]

(A) $\dfrac{2}{700}$ (B) $\dfrac{2}{350}$
(C) $\dfrac{3}{350}$ (D) $\dfrac{3}{700}$

Q38. A gaseous mixture of propane and butane of volume 3 litre on complete combustion produces 11.0 litre CO_2 under standard condition of temperature and pressure. The ratio of volume of butane to propane is ?

(A) 1 : 2 (B) 2 : 1
(C) 3 : 2 (D) 3 : 1

Q39. $C_6H_5OH\,(g) + O_2\,(g) \longrightarrow CO_2(g) + H_2O\,(l)$
Magnitude of volume change if 30 ml of C_6H_5OH (g) is burnt with excess amount of oxygen, is-

(A) 30 ml (B) 60 ml
(C) 20 ml (D) 10 ml

Q40. A mixture of C_2H_2 and C_3H_8 occupied a certain volume at 80 mm Hg. The mixture was completely burnt to CO_2 and H_2O(l). The pressure of CO_2 was found to be 230 mm Hg at the same temperature and volume. The mole fraction of C_2H_2 in mixture is?

(A) 0.125 (B) 0.5 (C) 0.85 (D) 0.25

Q41. Which is similar for 4g CH_4 and 20 gram SO_3 At constant temperature and pressure

(A) Molecular mass (B) Density
(C) Volume (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Atomic mass of elements are defined with respect to $1/12^{th}$ of mass of single atom of C-12 [present scale]. If reference is changed to $1/24^{th}$ of mass of single atom of C-12 [new scale]. Then select the correct statement(s).

Given : Atomic mass of Fe on present scale in 56

Mass of $1/12^{th}$ of mass of single atom of C-12 =1amu

Mass of $1/24^{th}$ of mass of single atom of C-12 =1uma

(A) Atomic mass of elements will change

(B) Mass of an atom (in gm) of an element remain same in both scale

(C) Mass of an atom of Fe will be 112 amu on new scale.

(D) Atomic mass of Fe on new scale will be 112.

Q2. Choose the correct pair of empirical formula of oxides of nitrogen which contains 7 g of nitrogen in their 15 g, 23g or 19 g.

(A) NO & N_2O_3 (B) NO & NO_2

(C) N_2O_3 & N_2O_5 (D) NO_2 & N_2O

Q3. 20 ml of a gaseous compound containing nitrogen and hydrogen only, is passed through a series of electric spark, by which the volume becomes 40 ml. When the resulting gases are passed through water, the volume becomes 30 ml. Now, 50 ml oxygen gas is mixed in the resulting gases and fired. If the final volume becomes 50 ml, the correct information(s) is/are (All the volumes are measured at the same temperature and pressure). N_2 and H_2 gases are completely insoluble in water)

(A) The gaseous compound is NH_3

(B) The gaseous compound is N_2H_4

(C) Only 50% of the gaseous compound was decomposed into N_2 and H_2

(D) 100 % of the gaseous compound was decomposed into N_2 and H_2

Q4. Consider the following chemical reaction $Pb(NO_3)_2 + Na_2SO_4 \rightarrow PbSO_4 + 2NaNO_3$. If a series of experiments are run maintaining sum of the weights of two reactant consant but varying the weights of reactants which of the following statements is (are) true ?

$[M_{Pb(NO3)2} = 394, M_{Na2SO4} = 142]$

(A) Maximum weight of the ppt ($PbSO_4$) will be formed if equal weights of reactants are taken.

(B) Maximum weight of the ppt ($PbSO_4$) will be formed if equal moles of reactants are taken.

(C) In the experiment as the weight of Na_2SO_4 increases weight of ppt($PbSO_4$) increases

(D) In the experiment as the weight of $Pb(NO_3)_2$ increases weight of ppt($PbSO_4$) increases

Q5. 2.2 g CO_2 has same number of atoms of oxygen as in [At. wt. of C = 12, N = 14, O = 16]

(A) 0.05 mole NO_2

(B) 0.025 mole H_2SO_4

(C) 50 millimoles SO_2

(D) 2.2 g N_2O

Q6. A sample of chalk ($CaCO_3$) contains clay as impurity. The clay loses 11% of its weight as moisture on prolong heating. A given 5g sample of chalk on heating shows a loss in weight (due to evolution of CO_2 (from $CaCO_3$) and water) by 1.1 g. Then.

(A) % by weight of $CaCO_3$ in sample is 33.4%

(B) % by weight of CO_2 in weight loss is 73.48 %

(C) Weight of clay remaining after losing moisture is 3.33 g

(D) % by weight of CO_2 in wt. loss is 33.33%.

Q7. Equal weight of 'X' (At. wt. = 36) and 'Y' (At. wt. = 24) are reacted to from the compound X_2Y_3. Then

(A) X is the limiting reagent

(B) Y is the limiting reagent

(C) No reactant is left over

(D) Mass of X_2Y_3 formed is double the mass of 'X' taken

Q8. 0.2 mole CO_2 gas is absorbed completely in 500 ml NaOH solution. The absorption is only due to reaction of CO_2 with NaOH to form Na_2CO_3 or $NaHCO_3$ or both. The correct information(s) is/are ?

(A) The minimum molarity for NaOH solution is 0.4 M.

(B) If no NaOH is left after reaction, the maximum molarity of NaOH solution is 0.8 M.

(C) Equal moles of $NaHCO_3$ and Na_2CO_3 will form, if concentration of NaOH solution is 0.6M.

(D) If concentration of NaOH solution is 0.9M, 0.3 mole NaOH will left unreacted.

Q9. The **incorrect** statement(s) regarding 2M $MgCl_2$ aqueous solution is/are :

($d_{solution}$ = 1.09 g/mL)

(A) Molality of Cl^- is 4.44 m

(B) Mole fraction of $MgCl_2$ is exactly 0.035

(C) The conc. of $MgCl_2$ is 19% w/v

(D) The conc. of $MgCl_2$ is 19×10^4 ppm

Q10. 25 moles of mixture of SO_2 & O_2 was passed over a catalyst 8 moles of SO_3 was formed. After reaction the final mixture composition is/are -

(A) 9 mole of O_2, 8 mole of SO_3

(B) 13 mole of SO_2, 8 mole of SO_3

(C) 9 mole of O_2, 12 mole of SO_3

(D) 15 mole of O_2, 10 mole of SO_3

Q11. A 5L vessel contains 2.8 g of N_2. When heated to 1800 K 30% molecules are dissociated into atoms :

(A) total number of moles of atoms in the container will be 0.06

(B) total number of molecules in the container will be 0.07 N_A

(C) total number of moles of atoms and molecules in the container will be 0.13

(D) total number of moles of atoms in the container will be 0.07

Q12. 400 g mixture of AB and AC_2 contains 26.25% of "A" by mass. Select the correct statement(s).

(Atomic masses of A = 15, B = 25, C = 22.5)

(A) % mass of AB is 10

(B) Number of moles of AC_2 in the mixture is 4

(C) Number of moles of AB is 1

(D) Total moles of "A atoms" in mixture is 7

Q13. 50 ml. of CO is mixed with 20 ml of oxygen and sparked. After reaction, the mixture is treated with an aqueous KOH solution. Choose the correct option -

(A) The volume of the CO that reacts is 40ml

(B) The volume of the CO_2 formed is 40 ml

(C) The volume of the CO that remains after treatment with KOH is 20ml.

(D) The volume of mixture obtained after reaction when treated with KOH solvent remains 10 ml.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q.14 to 15

3.00 g of an organic compound containing C, H, O, S and N only, is burnt completely by excess of oxygen gas. The gaseous products is first cooled to room temperature, by which water vapour is condensed and absorbed completely by anhydrous $CaCl_2$. The mass of salt increased by 2.16 g.

Now, the remaining gases were passed through excess of aqueous NaOH solution by which the mass of solution increased by 5.68 g. The gases remained, were passed through excess alkaline pyragallol solution. The gas still present occupied 150 ml at 1.642 atm and 27°C.

In an another experiment, 6.00 g of same organic compound produced 9.32 g $BaSO_4$, after treatment of series of non-sulphur reagents (Ba = 137)

Q14. Which of the following is correct information about the percentage composition of the compound, by mass ?

(A) Hydrogen = 8 %

(B) Carbon = 51.64 %

(C) Sulphur = 21.33 %

(D) Nitrogen = 9.33 %

Q15. On combustion , the mole ratio of CO_2 and SO_2 formed is?

(A) 1 : 1 (B) 5 : 2

(C) 5 : 1 (D) 3 : 2

Paragraph for Q.16 to 18

Molarity, molality, mole fraction are used in these days for stoichiometric calculations. It is necessary to write balanced chemical equation when calculations are done in terms of molarity.

Q16. 38% HCl has density equal to 1.20 g ml^{-1}. The molality and molarity respectively are?

(A) 12.4, 16.7 (B) 16.7, 12.4

(C) 12.4, 12.2 (D) 16.7, 16.7

Q17. 10 ml of 1M $BaCl_2$ solution, 5 ml of 0.5 M K_2SO_4 is added $BaSO_4$ is precipitated out. The amount of $BaSO_4$ ppt. will be ?

(A) 0.0025 moles (B) 0.025 moles

(C) 0.00025 moles (D) 0.0050 moles

Q18. Mole fraction of solute in aqueous solution is 0.2. The molarity of solution will be ?

(A) 13.88 (B) 1.388

(C) 0.138 (D) 0.0138

Paragraph for Q.19 to 20

Questions given below are based on two industrial observations :

Phosphoric acid H_3PO_4, in widely used to make fertiliser & can be prepared is a two step process

Step I : $P_4 + 5O_2 \rightarrow P_4O_{10}$

Step II : $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$

We allow 310 grams of phosphorus to react with excess oxygen, which forms tetraphosphorous decoxide P_4O_{10}, in 50 % yield. In step II reaction 25 % yield of H_3PO_4 is obtained.

Q19. Assume 100 % yield, then number of moles of H_3PO_4 would be obtained –

(A) 1.25 (B) 10.0

(C) 5.0 (D) 2.5

Q20. Actual number of mole of H_3PO_4 obtained in experiment conditions is?

(A) 5 (B) 2.5

(C) 1.25 (D) 0.3125

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q21. **List-I**

(P) 1.2 g of Mg

(Q) Molarity of 35.5 g chlorine in 0.5 litre water

(R) Density of solution is 10x g/ml

(S) Strength of a solution is 1000x g in one litre

List-II

(1) Hence 1x gm of solute is dissolved in 1 ml olution

(2) One ml of solution weighs 10x g

(3) Reacts with 0.8 g of oxygen to form 2.0 g of MgO

(4) 1 M

Code :

	P	Q	R	S
(A)	3	4	1	2
(B)	2	4	3	1
(C)	3	4	2	1
(D)	3	2	4	1

Q22. **List-I(solvent)**

(P) Turpentine oil

(Q) $CuSO_4/CaCl_2$

(R) KOH

(S) Alkaline pyrogallol

List-II (gases absorbed)

(1) H_2O

(2) O_2

(3) CO_2, SO_2,Cl_2

(4) O_3

Code :

	P	Q	R	S
(A)	3	2	1	2
(B)	2	4	1	3
(C)	4	1	3	2
(D)	2	3	1	4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q23. **Match the column :**

Column-I

(A) 6.3 g HNO_3 in 100 kg of solution

(B) 4 g MgO and 360 g water

(C) 2M H_2SO_4

(D) 40% $\left(\frac{w}{w}\right)$ NaOH solution ($d_{solution}$ = 1.2 g/mL)

Column-II

(P) $X_{solvent} = \frac{200}{201}$ (X = mole fraction)

(Q) $\% \frac{w}{v} = 19.6$

(R) 12 M

(S) 0.277 m

(T) 63 ppm

Q24.

Column I	Column II
(A) 10 M MgO ($d_{solution}$ = 1.20 g/ml) Solute : MgO Solvent: H_2O	(P) $W_{solvent}$ = 120 g per 100 ml of solution
(B) 40% w/v NaOH ($d_{solution}$ = 1.6 g/ml) Solute : NaOH Solvent: H_2O	(Q) $W_{solution}$ = 150 g per 100 g solvent
(C) 8 m $CaCO_3$ Solute : $CaCO_3$ Solvent: H_2O	(R) W_{solute} = 120 g per 100 g of solvent
(D) 0.6 mol fraction of 'X' (molecular mass = 20) in 'Y' (molecular mass 25) Solute : X Solvent : Y	(S) $W_{solvent}$ = 125 g per 100 g of solute

Q25. Matching list :

Column-A		Column-B	
(A)	On decomposition 50% increase in volume at the same temperature and pressure	(i)	NH_3
(B)	On decomposition 100% increase in volume at the same temperature and pressure	(ii)	HCl
(C)	On decomposition no change in volume at the same temperature and pressure	(iii)	O_3
(D)	Gases highly soluble in water	(iv)	$COCl_2$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q26. A mixture of sodium chloride (NaCl) and anhydrous sodium carbonate (Na_2CO_3) has a mole ratio 2 : 1. It is dissolved in water and treated with $BaCl_2$ solution. The mass of $BaCO_3$ precipitated is 98.5 g. Calculate the mass **(in grams)** of NaCl in the mixture.

Q27. In a binary liquid solution, the mass percent of solute is $\frac{25}{7}$ and the ratio of mole fractions of solute and solvent is 1 : 9. If the ratio of molar masses of solute and solvent is 1 : x, then the value of 'x' is?

Q28. In one process for waterproofing, a fabric is exposed to $(CH_3)_2SiCl_2$ vapour. The vapour reacts with hydroxyl groups on the surface of the fabric or with traces of water to form the waterproofing film $[(CH_3)_2SiO]_n$, by the reaction

$$n(CH_3)_2SiCl_2 + 2nOH^- \longrightarrow 2nCl^- + nH_2O + [(CH_3)_2SiO]_n$$

where n stands for a large integer. The waterproofing film is deposited on the fabric layer upon layer. Each layer is 6.0 Å thick [the thickness of the $(CH_3)_2SiO$ group]. How much $(CH_3)_2\ SiCl_2$ is needed to waterproof one side of a piece of fabric, 1.00 m by 3.00 m, with a film 300 layers thick ? The density of the film is 1.0 g/cm^3.

Q29. When C_2H_4 is burnt in air, carbon dioxide and water are formed. If on combustion of C_2H_4, 2 mole of CO_2 is produced then calculate the number of drops of water produced along with this quantity of CO_2, if each drop contains 6.022×10^{21} water molecule.

Q30. Units of parts per million (ppm) or per billion (ppb) are often used to describe the concentrations of solutes in very dilute solutions. The units are defined as the number of grams of solute per million or per billion grams of solvent. Bay of Bengal has 1.9 ppm of lithium ions. What is the molality of Li^+ in this water ?

Q31. Titanium, which is used to make air plane engines and frames, can be obtained from titanium tetrachloride, which in turn is obtained from titanium oxide by the following process :

$3\ TiO_2(s) + 4C\ (s) + 6Cl_2\ (g) \longrightarrow 3TiCl_4(g) + 2CO_2(g) + 2CO\ (g)$

A vessel contains 4.32 g TiO_2, 5.76 g C and; 6.82 g Cl_2, suppose the reaction goes to completion as written, how many gram of $TiCl_4$ can be produced ? (Ti = 48)

Q32. When 100 ml of a $O_2 - O_3$ mixture was passed through turpentine, there was reduction of volume by 20 ml. If 100 ml of such a mixture is heated, what will be the increase in volume?

Q33. 1 gm sample of $KClO_3$ was heated under such conditions that a part of it decomposed according to the equation

(1) $2KClO_3 \rightarrow 2\ KCl + 3O_2$ and remaining underwent change according to the equation.

(2) $4KClO_3 \longrightarrow 3\ KClO_4 + KCl$

If the amount of O_2 evolved was 112 ml at 1 atm and 273 K., calculate the % by weight of $KClO_4$ in the residue.

Q34. 40 ml of a mixture of C_2H_2 and CO is mixed with 100 ml of O_2 gas and the mixture is exploded. The residual gases occupied 104 ml and when these are passed through KOH solution, the volume becomes 48 ml. All the volume are at same the temperature and pressure. Determine the composition of original mixture.

SUBJECTIVE TYPE

Q35. Preparation of Na_2SnO_2 involves the following set of reaction.

(i) $Sn + 2HCl \longrightarrow SnCl_2 + H_2$

(ii) $SnCl_2 + 2NaOH \longrightarrow Sn(OH)_2 + 2NaCl$

(iii) $Sn(OH)_2 + 2NaOH \longrightarrow Na_2SnO_2 + 2H_2O$

(a) Mass of Sn (in kg) required to produce 19.7kg Na_2SnO_2, assuming 100% yield of each reaction.

(b) If % yield of reaction (i), (ii), (iii) 25%, 50% , 40% respectively. Calculate the mass of Sn (in kg) required to produce 19.7kg of Na_2SnO_2.

Q36. A sample of fuming sulphuric acid containing H_2SO_4, SO_3 and SO_2 weighing 1450 g is found to require 18 L of 2.00 M alkali (NaOH) for neutralisation. A separate sample shows the presence of 5 mole SO_2. Find the percentage of "free" SO_3, H_2SO_4 and "combined" SO_3in the sample.

Q37. A mixture of H_2 and butene (C_4H_8) was collected in a Eudiometer tube. Then, 80 ml of oxygen were also introduced. The resulting mixture of all the gases was exploded. **After** cooling **resulting gaseous mixture passes through** KOH solution which caused a contraction of 32 ml and 16ml of oxygen alone were left behind.

(a) Calculate total volume of initial mixture.

(b) Calculate percentage volume of butene in original gas mixture.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	A	C	C	C	C	A	D	D	D
11	12	13	14	15	16	17	18	19	20
B	D	C	B	B	C	A	A	C	B
21	22	23	24	25	26	27	28	29	30
A	D	A	B	B	B	D	C	C	A
31	32	33	34	35	36	37	38	39	40
B	D	C	A	B	A	A	A	B	A
41	42	43	44	45	46	47	48	49	50
D	A	B	D	C	C	B	A	C	C
51	52	53	54	55	56	57	58	59	60
D	B	B	A	B	D	B	C	B	C
61	62	63	64	65	66	67	68	69	70
C	C	C	B	B	D	B	D	B	A
71	72	73	74	75	76	77	78	79	80
C	A	C	B	A	B	D	B	D	D
81	82	83	84	85	86	87	88		
A	A	B	D	A	C	C	A		

Ex-1A

1. A,C,D **2.** B,C,D **3.** D **4.** A,B,C **5.** A, C
6. [B,C] **7.** B,D **8.** A,C **9.** A,B,C,D **10.** B,D
11. A,B,D **12.** [A,B] **13.** A,B,D **14.** B,C,D **15.** A,D
16. A,C **17.** A,B,C **18.** A,C **19.** A,B,C **20.** A
21. C **22.** B **23.** D **24.** A **25.** C
26. D **27.** A **28.** B,D **29.** A **30.** C
31. D **32.** C **33.** C **34.** D **35.** B
36. C **37.** A **38.** C **39.** A
40. A → Q; B → P,R ; C → P,R,T; D →P,R,T
41. A → P,R,S;B → P,Q,R,S ; C → Q; D →P, **42.** C
43. (A) → P,T ; (B) → Q ; (C) → R ; (D) → P, S **44.** 8 **45.** 8,2
46. 5 **47.** 13×10^{-1} **48.** 2M **49.** 1.288 **50.** 1.736 litre
51. 5 ml, 2 ml, 3 ml **52.** C_2H_2 = 6 ml, CO = 14 ml **53.** 30 ml
54. 2 **55.** B **56.** D
57. D **58.** (a) 0.169; (b) 118 %
59. (a) CO_2, H_2O and O_2

(b) $n_{CO_2} = 3$, $n_{H_2O} = 4$;

(c) C_2H_4 and CH_4 are the H.C

(d) Total Volume = (3+2)mol O_2 & 2 mol C_2H_4 & 1 mol CH_4 = 8 mol

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	D	A	D	A	D	C	C	C	D
11	12	13	14	15	16	17	18	19	20
A	A	C	B	D	B	C	D	C	C
21	22	23	24	25	26	27	28	29	30
B	A	C	A	A	B	A	A	B	B
31	32	33	34	35	36	37	38	39	40
A	D	D	A	C	C	A	B	B	A
41									
C									

Ex-2A

1. A, B, D **2.** A,B **3.** B,C **4.** B,D **5.** A,B,C

6. A,B,C **7.** C,D **8.** A,B,C **9.** B,D **10.** A,B

11. A,B,C **12.** A,C,D **13.** A,B,D **14.** A,C,D **15.** C

16. B **17.** A **18.** A **19.** B **20.** C

21. C **22.** C

23. (A) → (T) ; (B) → (P,S) ; (C) → (Q) ; (D) → (R)

24. (A) → (Q); (B) → (P); (C) → (S); (D) → (R)

25. (A) → (iii), (B) → (i, iv) , (C) → (ii) , (D) → (i, ii, iv)]

26. 58.5 **27.** 3 **28.** [0.9413 g] **29.** 200 **30.** 2.7×10^{-4}

31. 9.12 **32.** 10 ml **33.** [59.72%] **34.** C_2H_2 = 16 ml, CO = 24 ml

35. (a) 11.9 kg (b) 238 kg

36. H_2SO_4 = 35.38%, Free SO_3 = 63.1%, combined SO_3 = 28.89%

37. (a) 40 ml, (b) 20%